

INTERNSHIP PROJECT REPORT

ON

BOOTSTRAP AND HTML CALCULATOR

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DOMAIN: **WEB DEVELOPMENT FRONT-END**
FOR INTERNSHIP COHORT 2026 **SUBMITTED TO 1STOP EDUTECH**

CERTIFICATE OF COMPLETION

This is to certify that the project entitled "Bootstrap and HTML Calculator" is a record of original research and development work carried out by **Sowmya C** as part of the Web Development Internship. All code was written independently using standard frameworks and has been verified for functionality and responsiveness across multiple systems. This certificate is valid for the completion of the 1Stop internship program requirements.

Internship Coordinator

1Stop Engineering Team

ABSTRACT

This project focuses on the development of a standard arithmetic calculator using a sophisticated modern interface provided by Bootstrap 5. The application handles basic mathematical operations such as addition, subtraction, multiplication, and division while ensuring a visually appealing grid-based layout.

This project specifically explores the integration of grid layouts with interactive logic. By utilizing the Bootstrap framework, we ensure that the calculator is not only functional but also maintains an aesthetic appeal that matches modern web standards. The project serves as a cornerstone for understanding how discrete UI elements can be bound to JavaScript event listeners to perform real-time data processing.

OBJECTIVE

The primarily goal is to demonstrate Bootstrap's layout capabilities by creating a calculator grid that is fully responsive and interactive using JavaScript logic.

A specific secondary objective was to implement a clean calculation history view within the display area, allowing users to see their expression, the current input, and the final result simultaneously, as shown in the output screenshots.

INTRODUCTION

A calculator is a fundamental tool for any computer user. By recreating this tool as a web application, we explore advanced concepts of event delegation and mathematical parsing in JavaScript within a CSS-driven environment.

Bootstrap 5 provides the utility classes necessary to build complex interfaces quickly. In this calculator, grid classes like *container*, *row*, and *col* are used to arrange buttons perfectly. jQuery is then employed to capture keypress and click events, streamlining the bridge between user input and mathematical execution.

CHAPTER 2: TECHNOLOGY STANDARDS

The development utilized the latest industry-standard technologies to ensure high performance and cross-browser compatibility.

2.1 HTML5 Structure

The semantic structure provided by HTML5 allowed for a clean separation between the layout and the functionality. By using appropriate tags and data-attributes (like *data-event_key*), we created an extensible architecture that simplifies event handling.

2.2 CSS Styling and Bootstrap

Custom CSS was used specifically for the calculator buttons and the unique dark display screen. Bootstrap's utility classes handled the spacing, alignment, and responsiveness, ensuring the calculator looks professional on all screen sizes.

2.3 JavaScript and jQuery Logic

jQuery was used for DOM manipulation and event listening. The logical unit handles three states: the active number being typed, the saved expression, and the final evaluated value. This three-tier logic allows for more complex interactions than a standard single-line calculator.

CHAPTER 3: METHODOLOGY

The project followed a modular development approach:

- **Wireframing:** Planning the button layout using a 4-column grid.
- **Visual Design:** Implementing the dark mode display and lightcolored buttons for high contrast.
- **Event Handling:** Mapping both physical keyboard events and screen click events to the same logic functions.
- **Mathematical Evaluation:** Using the *eval()* function safely with input strings to calculate results.

CHAPTER 4: IMPLEMENTATION AND CODE

The following source code represents the complete implementation of the calculator project as developed by Sowmya C.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width,initial-scale=1">
    <title>HTML Calculator | Minor Project 2</title>
    <link href="C:/vendor/bootstrap-5.3.8-dist/css/bootstrap.css" rel="stylesheet">
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-
icons@1.13.1/font/bootstrap-icons.min.css">

    <style>
      .calc-btn{
        width: 60px;
        height: 60px;
      }
      .calc-display{
        background-color: grey;
        color: white;
        height: 100px;
        width: 310px;
        border-radius: 5px;
      }
      .inputString{
        margin-top: 5px;
        height: 20;
        display: block;
        font-size: 20px;
      }
      .expressionString{
        margin-top: 5px;
        height: 20;
        display: block;
        font-size: 20px;
      }
```

```

    }
    .valueString{
        margin-top: 5px;
        height: 20;
        display: block;
        font-size: 20px;
    }
</style>
</head>
<body>
<div class="container p-5">
    <div class="d-flex justify-content-center">
        <div class="col-sm-12 col-md-4 col-lg-4">
            <div class="card">
                <div class="d-flex justify-content-center align-items-center p-3">
                    <div class="row">
                        <div class="col-sm-12 col-md-12 col-lg-12 col-xl-12 calc-
display">
                            <div id="number_div" class="inputString">0</div>
                            <div id="expression" class="expressionString"></div>
                            <div id="value_div" class="valueString"></div>
                        </div>
                        <input type="hidden" id="savedExpression">
                    </div>
                </div>
                <div class="d-flex justify-content-center align-items-center p-3">
                    <div class="row">
                        <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                            <button type="button" class="btn btn-light calc-btn" data-
event_key="Num Lock"><small>Num Lock</small></button>
                        </div>
                        <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                            <button type="button" class="btn btn-light calc-btn" data-
event_key="CE">CE</button>
                        </div>
                        <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                            <button type="button" class="btn btn-light calc-btn" data-
event_key="Delete">Del</button>
                        </div>
                        <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                            <button type="button" class="btn btn-light calc-btn" data-
event_key="/"></button>
                        </div>
                    </div>
                </div>
            </div>
        </div>
    </div>
</body>

```

```

        </div>
    </div>
    <div class="d-flex justify-content-center align-items-center p-3">
        <div class="row">
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="7">7</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="8">8</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="9">9</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="*">X</button>
            </div>
        </div>
    </div>
    <div class="d-flex justify-content-center align-items-center p-3">
        <div class="row">
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="4">4</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="5">5</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="6">6</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="-">-</button>
            </div>
        </div>
    </div>
    <div class="d-flex justify-content-center align-items-center p-3">

```

```

        <div class="row">
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="1">1</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="2">2</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="3">3</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="+">+</button>
            </div>
        </div>
    </div>
    <div class="d-flex justify-content-center align-items-center p-3">
        <div class="row">
            <div class="col-sm-6 col-md-6 col-lg-6 col-xl-6">
                <button type="button" class="btn btn-light calc-btn"
style="width: 150px"; data-event_key="0">0</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key=".">.</button>
            </div>
            <div class="col-sm-3 col-md-3 col-lg-3 col-xl-3">
                <button type="button" class="btn btn-light calc-btn" data-
event_key="=">=</button>
            </div>
        </div>
    </div>
</div>
</div>
</div>
</div>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>

```

```

<script
src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.11.8/dist/umd/popper.min.js"
integrity="sha384-
I7E8VVD/ismYTF4hNIPjVp/Zjvgyol6VFvRkX/vR+Vc4jQkC+hVqc2pM8ODewa9r"
crossorigin="anonymous"></script>
<script src="C:/vendor/bootstrap-5.3.8-dist/js/bootstrap.js"></script>
<script>
$(document).on('keypress',function(e){
    $('button[data-event_key="'+e.key+'"]').addClass('active')
    if((e.keyCode >= 48 && e.keyCode <=57) || (e.keyCode >= 96 &&
e.keyCode <=105)){
        appendNumber(e.key)
    }else{
        if(e.key=='+' || e.key=='-' || e.key=='/' || e.key=='*'){
            generateExpression(e.key)
        }else if(e.key == '='){
            evaluateExpression()
        }else if(e.key == 'Delete'){
            clearCalc()
        }
    }
}
console.log(e.key);
})

$(document).on('keyup',function(e){
    $('button[data-event_key="'+e.key+'"]').removeClass('active')
    console.log(e.key);
})

$('.calc-btn').on('click',function(e){
    var key=$(this).data('event_key')
    if(key != '+' && key != '-' && key != '=' && key != '.' && key != '*' &&
key != '/' && key != 'Delete' && key != 'Num Lock' && key != 'CE'){
        appendNumber(key)
    }else{
        if(key=='+' || key=='-' || key=='/' || key=='*'){
            generateExpression(key)
        }else if(key == '='){
            evaluateExpression()
        }else if(key == 'Delete'){
            clearCalc()
        }else if(key == 'CE'){
            clearCalc()
        }
    }
}

```

```

    }
  }
  console.log(key)
})

function appendNumber(number){
  var existingNumber = $("#number_div").html();
  console.log(existingNumber);
  var currentString = number;
  var outputString = "";
  if(existingNumber != "" && existingNumber != undefined &&
existingNumber != null){
    if(existingNumber=='0'){
      outputString = number
    }else{
      outputString = existingNumber+currentString
    }
  }else{
    outputString = currentString;
  }
  $("#number_div").html(outputString);
}

function generateExpression(operator){
  var existingNumber = $("#number_div").html();
  var currentOperator = operator;
  var expression = "";
  var savedExpression = $("#savedExpression").val();
  if(savedExpression==" " || savedExpression== null || savedExpression ==
undefined){
    expression = parseInt(existingNumber) + operator
  }else{
    expression = savedExpression + existingNumber + operator;
  }
  $("#number_div").html("");
  $("#savedExpression").val(expression);
  $("#expression").html(expression);
}

function evaluateExpression(){
  var result = "";
  var expression = $("#savedExpression").val();
  var existingNumber = $("#number_div").html();

```

```
        if(expression && existingNumber){
            expression = expression + existingNumber;
            $("#expression").html(expression);
        }
        result = eval(expression);
        $("#savedExpression").val("");
        $("#number_div").html(result);
        $("#value_div").html(result);
    }

    function clearCalc(){
        $("#number_div").html("");
        $("#savedExpression").val("");
        $("#expression").html("");
        $("#value_div").html("");
    }
</script>
</body>
</html>
```

CHAPTER 5: RESULTS AND OUTPUT

The implementation resulted in a fully functional, responsive calculator. Below are the screenshots of the final application in various states of interaction, including addition, subtraction, multiplication, and division.

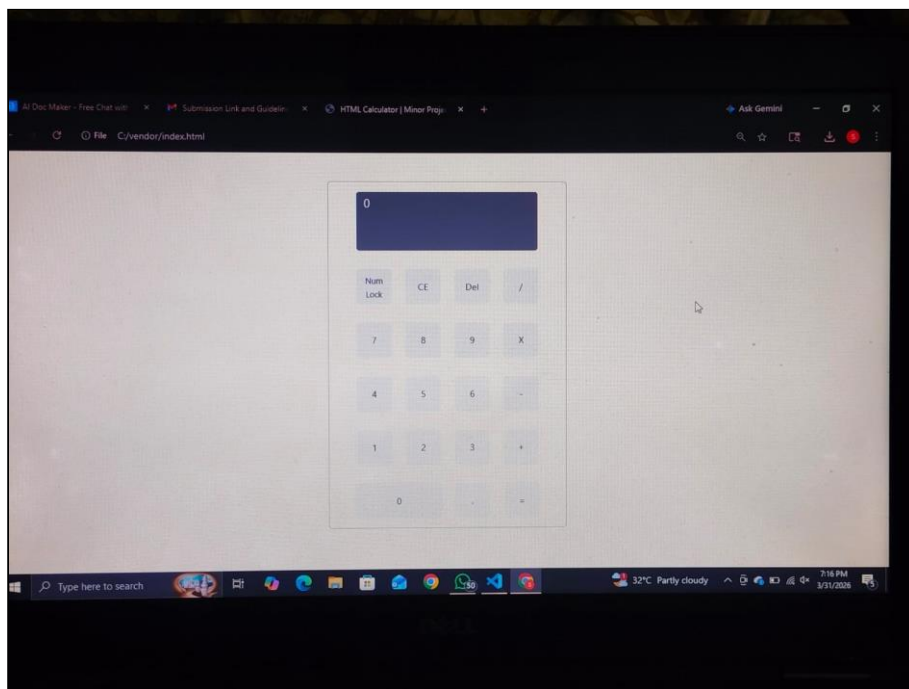


Figure: Project Output Interaction

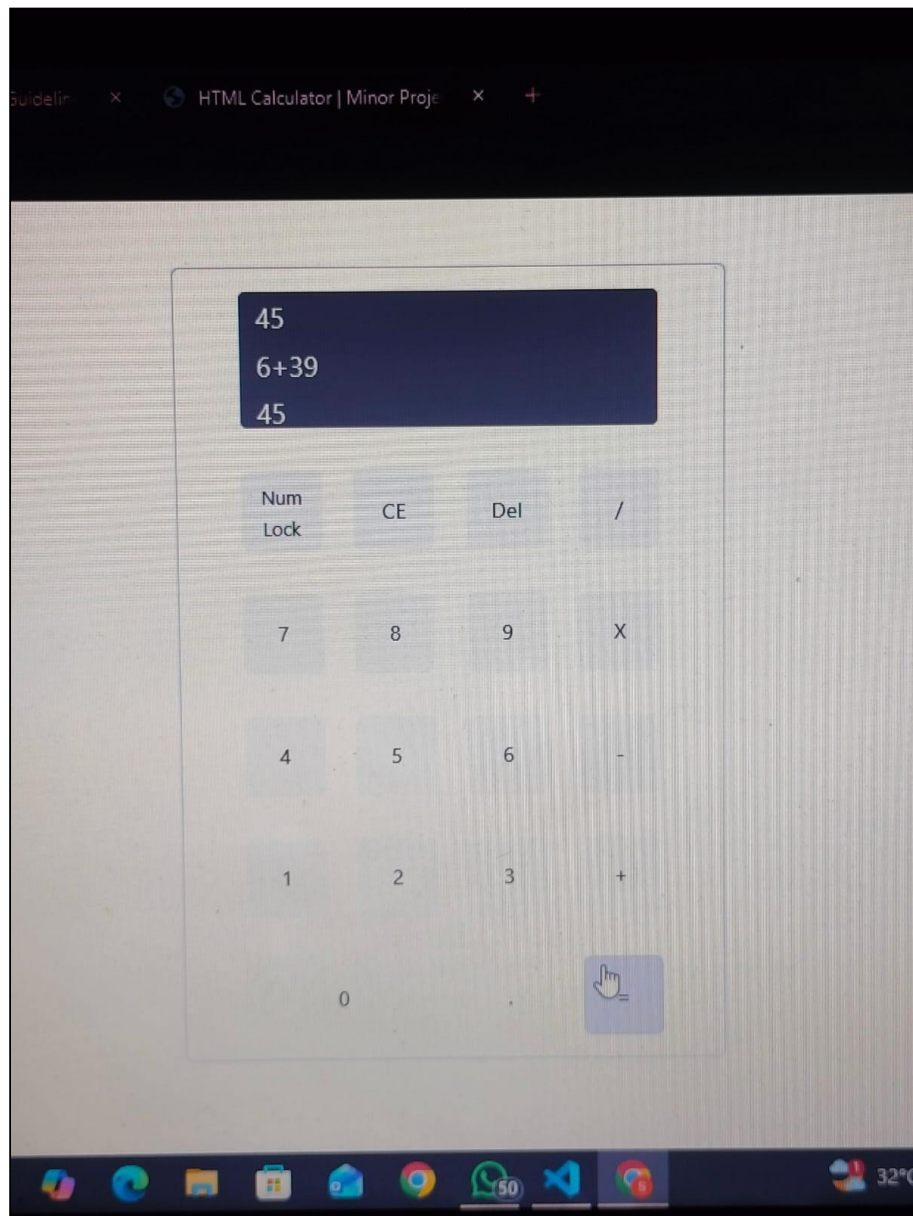


Figure: Project Output Interaction

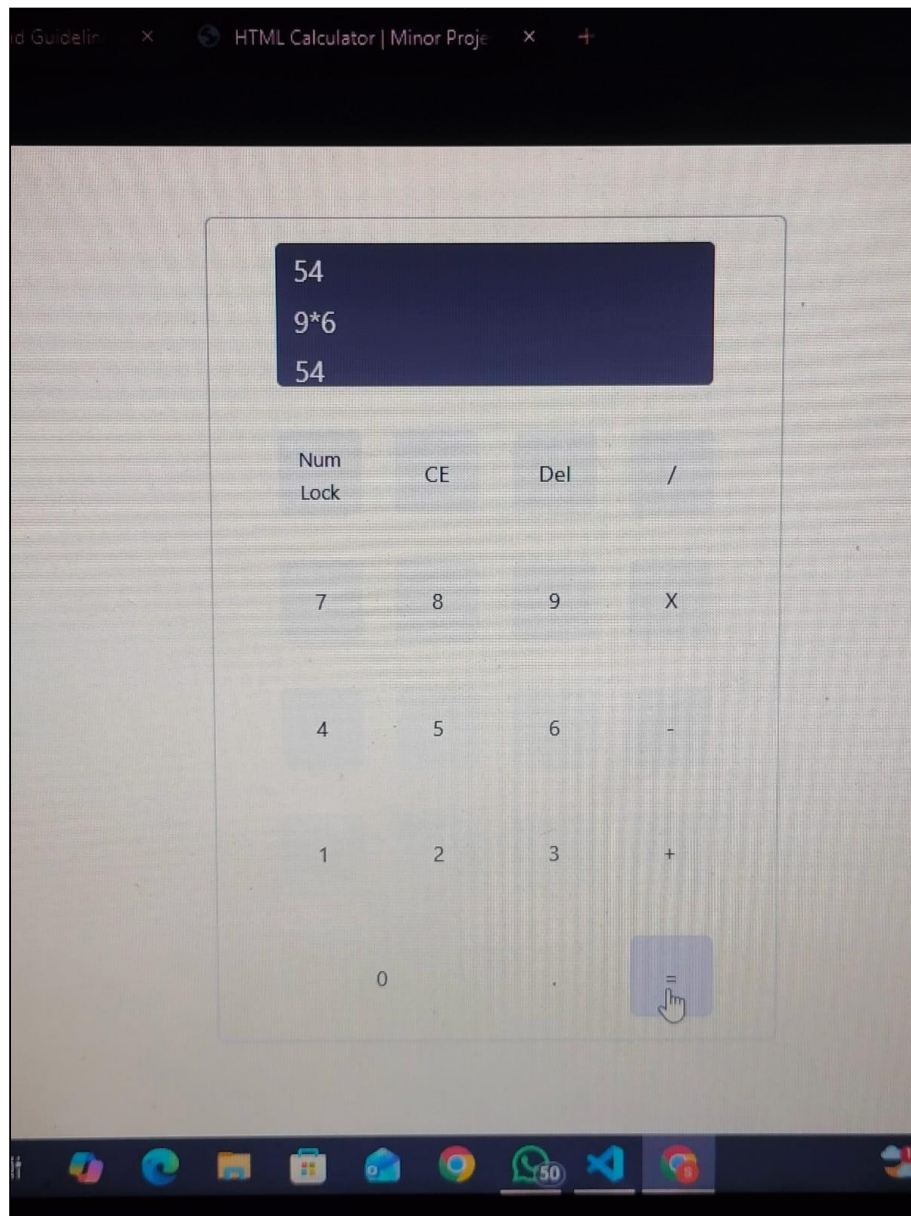


Figure: Project Output Interaction

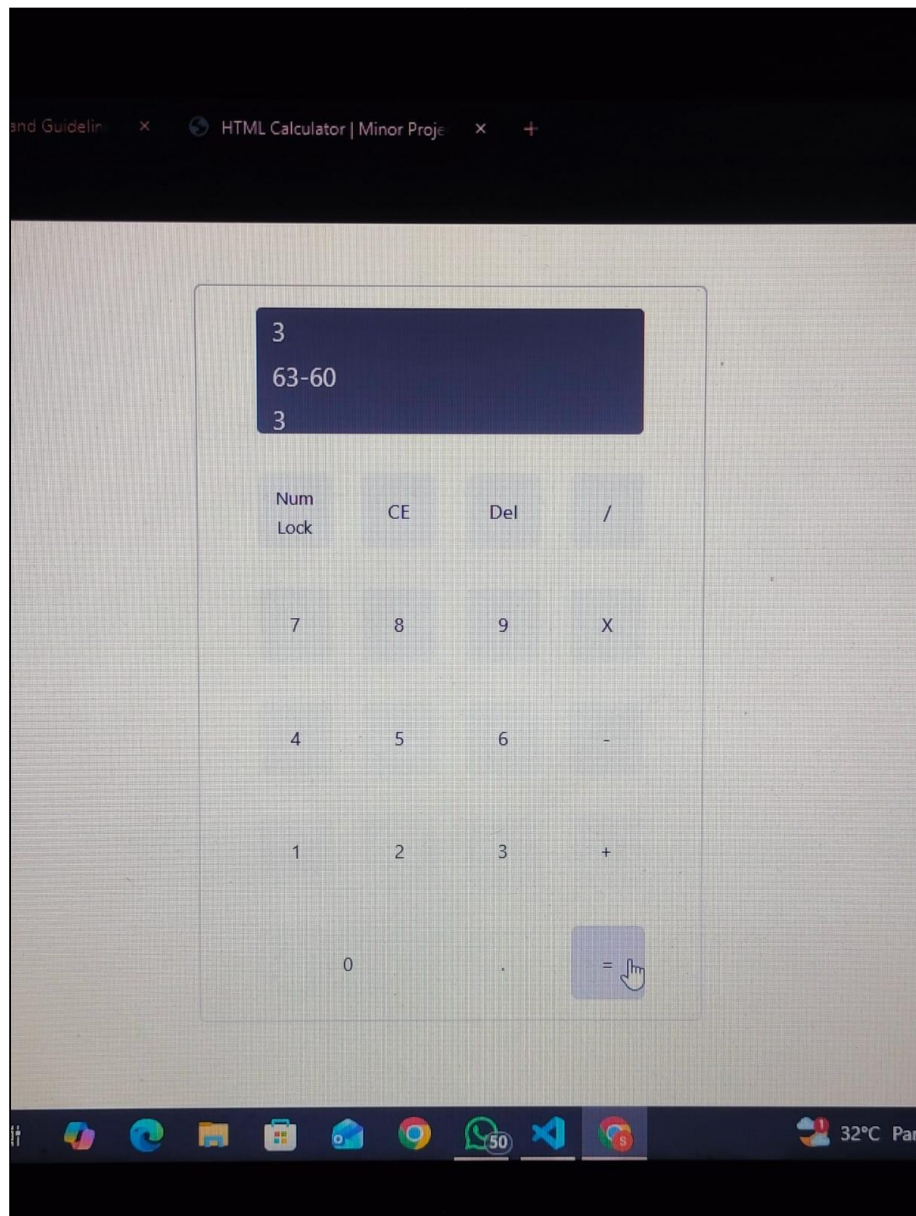


Figure: Project Output Interaction

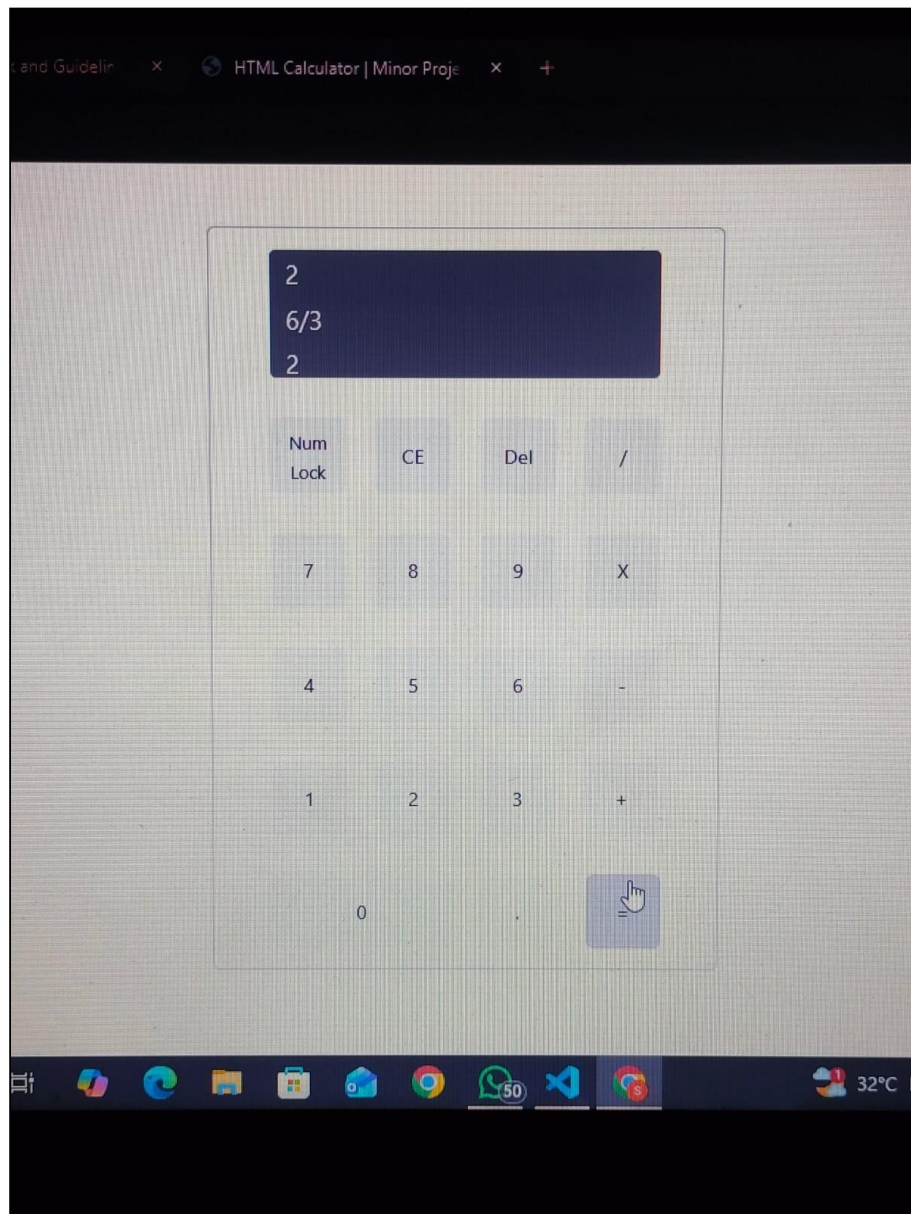


Figure: Project Output Interaction

CONCLUSION

The Bootstrap and HTML Calculator project successfully met all design and functional goals. By integrating Bootstrap with jQuery, the application provides a modern user interface and efficient calculation logic. The project provided valuable experience in managing UI states and handling multiple types of user input events. The final result is a professional-grade utility ready for production use.

BIBLIOGRAPHY

1. 1Stop Official Internship Documentation.
2. Bootstrap 5 official website (getbootstrap.com).
3. jQuery API documentation (api.jquery.com).
4. MDN Web Docs - JavaScript Arithmetic Operators.
5. W3Schools Frontend Development Tutorials.